



Recognising geometric sequences

1 _____ sequence is a sequence with a constant multiplicative relationship between successive terms. Tick **1** correct answer

☐ An arithmetic

☒ A geometric

☐ A multiplier

☐ A quadratic

2 In the geometric sequence 1.2, 4.8, 19.2, 76.8, ..., the common _____ between the terms is 4. Tick **2** correct answers

☐ difference

☒ multiplier

☒ ratio

☐ sequence

☐ term

3 Find the next two terms in this geometric sequence: 3, 30, 300, ... Tick **2** correct answers

☐ 330

☐ 600

☒ 3000

☒ 30 000

☐ 33 000



- 4 Which of these are terms in the geometric sequence generated by the rule: "Start on 224 and use a common ratio of $\frac{1}{4}$ "? Tick **3** correct answers

☒ 56☒ 14☐ $\frac{2}{7}$ ☒ $\frac{7}{8}$ ☐ 7.2

- 5 Some of these sequences are geometric, some are arithmetic. Select all the geometric sequences. Tick **3** correct answers

☐ 1, 2, 3, 4, ...☒ 1, 2, 4, 8, ...☒ 1000, 200, 40, 8, ...☐ 200, 80, -40, -160, ...☒ 1, 3, 9, 27, ...

- 6 Which statement is true of the sequence 1, 2, 6, 24, 120, ... ? Tick **1** correct answer

☐ It is geometric.☐ It is arithmetic.☐ It is geometric because the multipliers are: $\times 2$, $\times 3$, $\times 4$, $\times 5$ ☒ It has no common ratio therefore it is not geometric.